

Group Assignment

EBC4097 Quantitative Techniques for Financial Economics

Straddle strategy

Team

Nagesh Kellengre Prakash (i6418088)

Thodoris Papachronis (i6377299)

Tim Bruns (i6409478)

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1. Introduction

Market volatility is an essential concept in finance, as it provides traders and investors with insights into how much the price of an asset is expected to fluctuate. It describes how much the price of a market, index, or financial asset has changed or fluctuated over a given time period. This statistical metric, which reflects uncertainty or instability in investor mood, economic conditions, or external events, quantifies the frequency and intensity of price changes. It is the statistical indicator of how market index or financial instrument returns are distributed. It measures the level of risk or uncertainty connected to the size of price swings over a given time frame.

Greater and more erratic price fluctuations present investors with opportunities and difficulties; therefore, volatility is sometimes seen as a stand-in for risk. Although volatility is frequently thought of as a stand-in for risk, it also represents macroeconomic influences, investor psychology, and market dynamics. Fundamentally, supply and demand imbalances are the cause of volatility, which can be triggered by a variety of events, including the release of macroeconomic data (such as employment or inflation reports), geopolitical tensions (such as wars or trade disputes), corporate earnings surprises, changes in monetary policy (such as interest rate changes), or abrupt shifts in market sentiment (such as fear or greed).

There are two primary types of volatility: Historical volatility, which examines previous price movements, and Implied volatility, a forward-looking metric incorporated into options prices and represents traders' predictions of future price swings. While low volatility usually corresponds with stable, bullish markets where prices trend smoothly, high volatility sometimes occurs during uncertain times, such as economic recessions, pandemics, or political crises. A key factor in options pricing and, consequently, the effectiveness of tactics like the straddle is implied volatility (IV). Implied Volatility (IV) directly impacts option premiums and represents future market volatility predictions.

Actual volatility during a certain period is known as realized volatility, another important component of volatility. Calculating the standard deviation of an asset's returns over the relevant historical period yields the realized volatility. Realized volatility does not always indicate future price movements but provides insightful information about historical market activity. On the other hand, it helps compare the historical volatility of other assets or market circumstances and evaluate an asset's overall risk.

An excellent illustration of a volatility-driven options strategy that aims to capitalize on notable price fluctuations in either direction is the straddle strategy. Although it dates to the early days of options trading, its acceptance gained momentum when options pricing models were formalized. A straddle involves simultaneously buying a call option and a put option on the same underlying asset with identical expiration dates. For instance, if a stock ABC trades at 50 euros, a trader might buy a 50 euro call and 50 euro put expiring in one month. The strategy is directionally neutral; it does not bet on upward or downward movement but on the magnitude of movement. Figure 1 showcases the straddle strategy overview. The straddle profits if the asset's price moves significantly above or below the breakeven points, which are calculated as:

$$\begin{aligned}\text{Upper Breakeven} &= \text{Strike Price} + \text{Total Premium Paid} \\ \text{Lower Breakeven} &= \text{Strike Price} - \text{Total Premium Paid}\end{aligned}$$

If ABC's price surges to 60 or drops to 40, the trader profits from the call or put, respectively. If the price remains stagnant, both options expire worthless, and the trader loses the premiums paid.

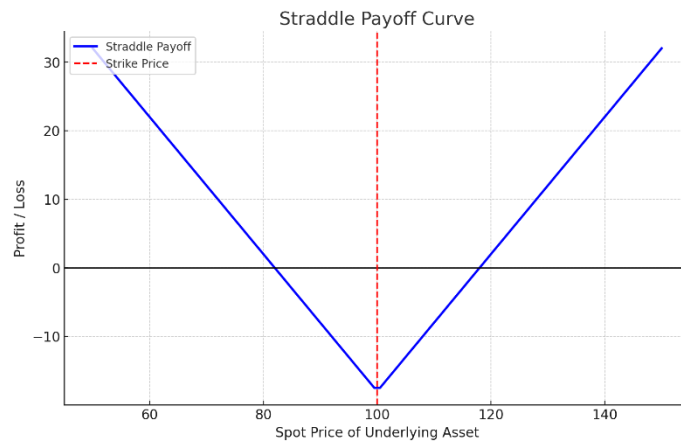


Figure 1 Straddle overview payoff curve

Forecasting volatility is an essential part of trading, especially for options traders who have to predict the state of the market. Econometric models such as Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models are used to forecast volatility. These models use historical price movements to predict future volatility and account for volatility clustering, which is the phenomenon where periods of high volatility are followed by more periods of high volatility. The Exponentially Weighted Moving Average (EWMA), which assigns greater weight to recent data, and stochastic volatility models, which represent volatility as a random process, are other forecasting methods. These models can be quite helpful in assessing implied volatility and modifying positions appropriately, particularly when using a volatility-heavy strategy like the straddle.

To sum up, volatility is a complex idea that has both historical and future aspects. Beyond merely comprehending historical asset price fluctuations, its importance lies in its ability to predict future market shifts and carry out successful trading plans. Finding the optimal timing to enter or exit the market and assessing the probability of good trade are critical tasks for the straddle strategy, which requires an awareness of and ability to estimate volatility.

2. Literature Review

This literature review provides an overview of the existing research on volatility, straddle strategy, and option pricing. Whaley (2000) notes that implied volatility is a vital input in pricing options and understanding market sentiment, as it indicates how much market participants expect the underlying asset to fluctuate. Engle (2001) asserts that methods such as GARCH models have become essential in econometrics for forecasting volatility and that volatility modeling is essential for comprehending and forecasting asset price movements.

Much research has been done on the difference between Implied and realized volatility. According to Poon & Granger (2003), Implied Volatility frequently overestimates future volatility, which presents straddle seller opportunities. However, Christensen & Prabhala (1998) demonstrated that implied volatility is more accurate than historical volatility at forecasting future volatility, particularly in relation to macroeconomic events.

For the straddle strategy to be successfully implemented, volatility forecasting is essential. Accurate projections of future volatility can greatly improve a trader's capacity to anticipate possible market movements, as the approach benefits from huge price swings. Econometric models like Engle's (2001) GARCH (Generalized Autoregressive Conditional Heteroskedasticity) model are frequently employed to forecast future volatility. The straddle strategy is more likely to be lucrative during extreme volatility, which these models are very good at predicting. By helping traders decide when to enter or leave deals, volatility forecasting approaches can improve the performance of trading strategies such as the straddle (Christoffersen, 2012).

According to Black and Scholes (1973), volatility is a key element of options pricing models, and higher volatility usually translates into higher option premiums. Buying a call and a put option on the same asset with the same strike price and expiration date is known as straddling, and it makes money when there are significant price swings in either direction because volatility directly affects the likelihood of large price fluctuations that would result in profits, it is crucial to the straddle strategy. The significance of volatility in options pricing is emphasized by Black and Scholes (1973), as it establishes the probability of an option expiring in the money. The Black-Scholes formula for the market volatility for straddle is

$$C = \Phi(d_1)S - \Phi(d_2)Ke^{-rT}$$
$$d_1 = \frac{\ln\left(\frac{S}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} \text{ and } d_2 = d_1 - \sigma\sqrt{T}$$

C = premium for the call option, Φ = cumulative density function of a standard normal distribution, T = expiration date, S = stock price at time t_0 , K = strike price, r = risk-free interest rate, and σ = volatility.

3. Methodology

The goal of this paper is twofold; first, different time series volatility models will be analyzed to find the optimal model for the returns of three different assets, Gold, as a commodity, Bitcoin, as a cryptocurrency, and finally Tesla, as a representative stock. The optimal model will then be used to generate the implied volatility which will be used as the basis for the high frequency straddle trading strategy. Here, the aim was to create a synthetic trading strategy that capitalizes on the implied volatility's advantages over more basic volatility models, such as the rolling volatility. From these goals, the following research questions were derived:

1. Which forecasting model is optimal in predicting the volatility of returns of assets?
2. How can we capitalize on volatility forecasting using a high frequency trading strategy?

To answer the first question, an analysis of different ARMA(p,q) (Hecq, A., & Velasquez-Gaviria, D. (2025)) and GARCH(p,q) (Gaviria, D. V. (2025)) models was performed based on the logarithmic returns of each asset. Daily closing price data from 01.01.2005 up to 06.03.2023 (the "training data") was considered for modeling purposes. The past two years were not considered here because they were later used to backtest the model and subsequently the trading strategy. It has to be noted that only trading days were considered. ARMA(p,q) and GARCH(p,q) models with $0 \leq (p, q) \leq 5$ were considered here. Information criteria were then used to compare all models, more specifically the Bayesian Information Criteria (BIC) with the one with the lowest BIC was deemed optimal.

The second research question was then analyzed using the previous findings. First, the optimal model was considered to forecast the implied volatility one day at a time for the timeframe of 07.03.2023 to 06.03.2025 (the "testing data"). Then 30 day rolling volatility was calculated for the testing data which was used 1) for the calculation of the synthetic options and 2) to as the basis for the training strategy. The expectation here is that previous literature findings of the optimal models, ARMA (1,0) and GARCH (1,1), for each asset, will be verified.

As there are no historic option prices available, synthetic options were constructed. The price of the options was calculated using the Black and Scholes formula. The expiration date T was 1 day (or 1/252), as this maximized the number of periods available for testing. A constant risk-free rate of annually 5% was used for simplicity. Using actual 10-year US treasury notes was considered, however, this would have increased the complexity of this analysis as US rates moved significantly during the testing period. As this paper aims to be proof of concept for the trading strategy, the added complexity would have only decreased the significance of this analysis. Also, a constant density function was assumed. The strike price K was always set at the money, so at the current asset price S. It is worth noting that a further analysis into an optimal K could increase the profitability of the strategy. Lastly, the rolling volatility was used for pricing of the options.

The straddle was "bought" every day, t, where the implied volatility of t+1 was greater than the rolling volatility of t. The idea of this trading strategy is that a higher implied volatility indicates a mispricing of the options which can be exploited. The returns of the straddle are calculated for every t when a straddle was bought, with the returns being equal to the sum of the cost of

both the call and put option and the payoff of each option. The payoff is defined as $\max(K - S, 0)$ for the call option and $\max(S - K, 0)$ for the put option.

The straddle trading strategy is expected to lose money most of the days when the straddle was bought and only offers infrequent returns, which should compensate for the losing periods. A clear clustering of straddle executions vs no execution of the strategy should be visible in the data, due to the tendency of assets to have volatility clustering.

4. Results

This section presents the empirical findings of the high frequency straddle strategy applied to TSLA, Bitcoin, and Gold. We evaluate the model performance through volatility and forecast. We will present the results from the ARMA and GARCH model for every asset. Also, we provide our results for this analysis as well as the limitations of the model.

4.1 Volatility Clustering

For this section, we are going to provide a graph that shows the strong volatility clustering among each of the three assets. According to the graph, Bitcoin has the largest returns swings, especially between 2020 and 2022. The reason for that was the pandemic of COVID-19 where the market was extremely volatile.

Tesla also showed a similar pattern, however not as volatile as Bitcoin. This could happen due to the economic impact of lockdowns on consumer demand and supply chains.

In contrast Gold remained relatively stable, experiencing less fluctuations than the other two assets. Hence it proves one more time that it is considered as a defensive asset, especially in stressful financial periods.

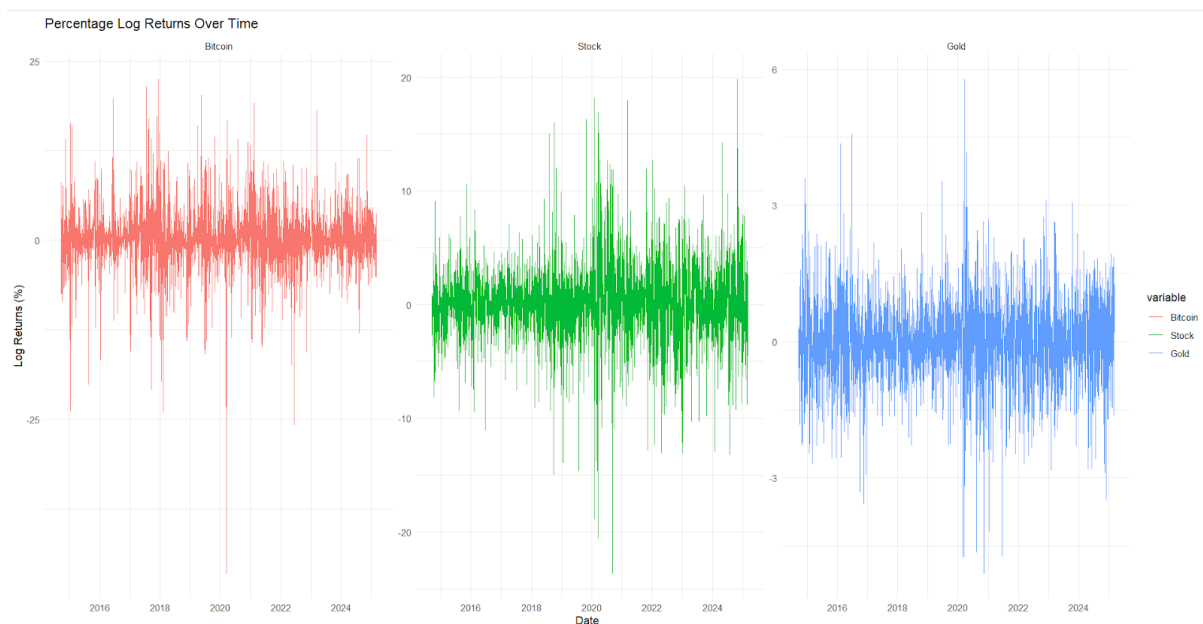


Figure 2. Log Return Volatility Clustering Across Assets (Bitcoin, TSLA, Gold)

4.2 Volatility Forecasting (GARCH Modelling)

In this section, we compare the forecasted volatility from a GARCH (1,1) model, which was the best model according to the calculations doing in R (see appendix for exact coefficients), with the 30-day rolling realized volatility for Bitcoin, TSLA, and Gold. We do that so we can understand the model captures volatility across each asset.

Bitcoin's large omega (ω) and alpha (α) show that there is a strong sensitivity to past shocks. However, the low β_1 makes it harder for GARCH to fully capture how it is going to behave in the future.

Tesla's stock has lower omega (ω) and alpha (α_1) than Bitcoin a very high β_1 (0.947). In other words, it has a longer memory, and it is not reacting that much to past shocks.

Gold on the other hand does not surprise much. Low alpha, omega and β_1 with low SD and kurtosis (3.52) making this asset perfect for defensive purposes.

The first graph below shows each asset's 30-day rolling standard deviation over the last 252 trading days. The GARCH forecasts (blue lines) follow the realized volatility (red dashed lines) reasonably well for Bitcoin and TSLA, especially when the market was considered unsustainable. However, the model cannot capture the sudden changes in Gold's performance mainly because it is considered a stable asset and does not provide any noise.

The three graphs in Appendix C display actual vs. forecasted log returns based on the ARMA-GARCH models. These graphs prove that GARCH can capture the general trend but underestimate sudden changes (spikes) in the market. Especially for Bitcoin, which is experiencing fat tails and high kurtosis. Same thing with Tesla but not as extreme as Bitcoin. On the other hand, the model for Gold performs really well, especially in stable periods, but struggles when the market conditions are more extreme.



Figure 3. 30-Day Rolling Volatility for Bitcoin, TSLA, and Gold (Last 252 Trading Days)

4.3 Return Forecasting (ARMA Models)

In this section we are going to focus on the ARMA models. We used Bayesian Information Criterion (BIC) to select the optimal ARMA models for each asset. The analysis showed that

for both Bitcoin and Tesla, the best model was ARMA (1,0) and for Gold the ARMA (0,1) (see appendix for exact coefficients).

Bitcoin is showing highly volatile behavior due to its nature. The ARMA (1,0) model captures short-term momentum during calm periods but fails during price jumps high BIC and high $\sigma^2=12.75$), which are frequent in Bitcoin.

Tesla has a relatively similar behavior to Bitcoin (high BIC = 11468.26, $\sigma^2 = 20.79$), especially when price announcements or other events affect the tech sector. Similar to Bitcoin cannot capture event-driven jumps leading to potential losses during these periods.

Gold in contrast shows with ARMA (0,1) and BIC = 5769.73, the best model in this analysis, a more predictable behavior. However, the transaction fees are way too high mostly due to low volatility of the asset leading to losses not because the strategy is wrong but because it is too expensive to be executed.

4.4 Strategy Profitability

The 1- day straddle is failing to produce significant and consistent profits across all three assets but the end of the backtesting period.

Gold experienced a downward trend throughout the entire period. Despite being a stable asset, the high costs for performing that strategy make it insufficient for generating profits.

Tesla is following a similar pattern even though we have some profits in 2025. Between 2023 and 2024 we lost a big portion of our capital, which seems that we are not going to take it badly.

Finally, Bitcoin presents the most important and interesting results. If we perform a straddle with this asset, we can clearly see that we have some periods with extremely big profits and some periods with losses. This leads to the fact that our strategy works only when the market is experiencing high volatility.

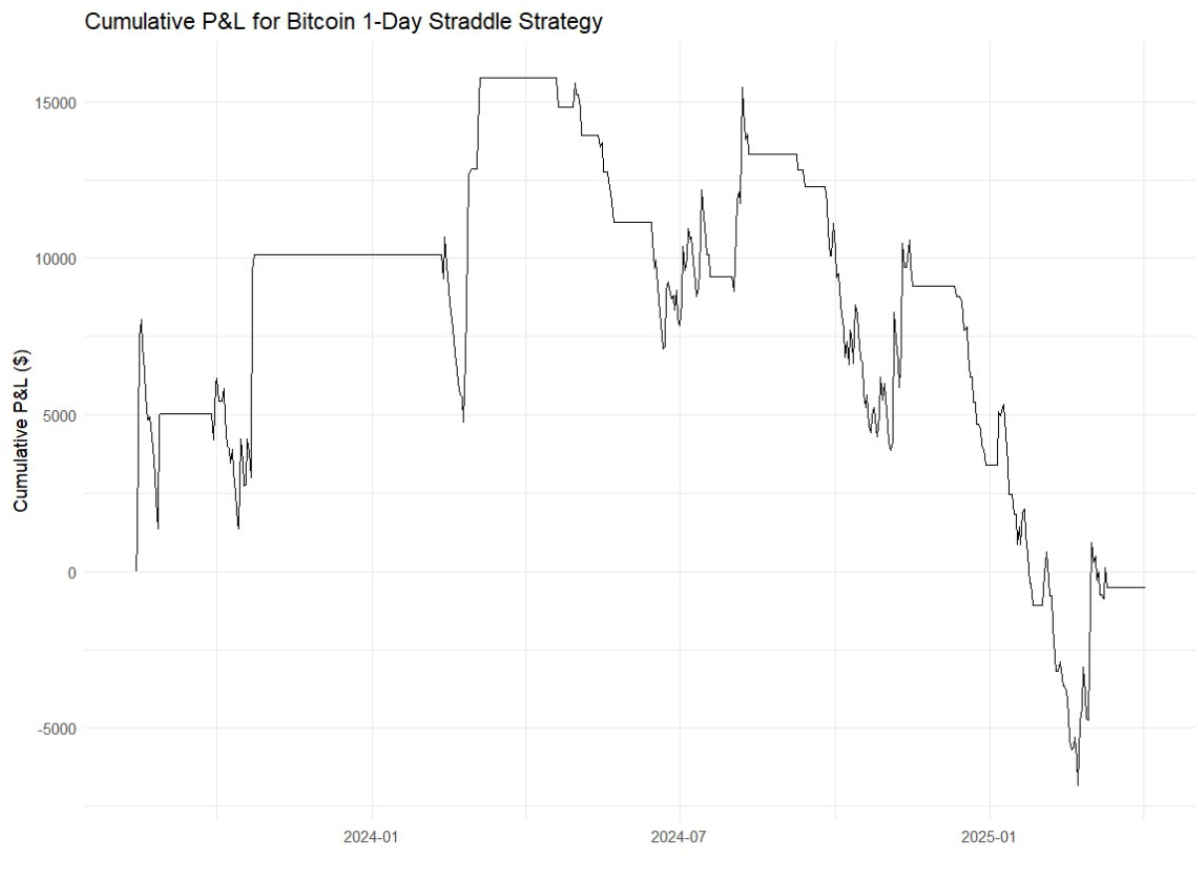


Figure 4. Cumulative P&L for Bitcoin 1-Day Straddle Strategy

The model should be adjusted for periods with low volatility to fit these conditions.

For high frequency trading (HFT) it is arguably more difficult to create a profitable strategy. However, the most effective way to do that is by adding some filters that prevent the model from executing a trade when the market conditions are not in our favor. Some of the filters can be applied when the market's volatility is or expected to be higher than we want. Likewise, the same principle can be used for transactions fees.

5. Conclusion

This paper tested a short-term straddle strategy on TSLA, Bitcoin, and Gold, using ARMA-GARCH models to forecast volatility and the Black-Scholes model for option pricing. A high volatility trading strategy was executed in periods where forecasted volatility exceeded realized volatility.

While the strategy is theoretically sound it under-performed when tested for real world conditions. Despite some periods with profits, if it is executed until the end, we will end up with losses among all three assets. The ARMA-GARCH models effectively captured general volatility trends but struggled to anticipate large shocks. Additionally, the options premium costs were too expensive, erasing our strategy's profits.

For future research, a straddle can still become profitable if it is being used and implemented with certain enhancements. These may include a Monte Carlo simulation using different market conditions for alternative scenarios. Also, a different volatility model might give a better picture of the market and help us make better predictions for the future.

6. References

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7. Appendix

Appendix A - GARCH coefficients and BIC:

	mu	omega	alpha1	beta1
	0.2071486	1.5049709	0.1440903	0.7974730
✓ Best GARCH model for Bitcoin: GARCH(1,1) with BIC = 5.7579				

	mu	omega	alpha1	beta1
	0.08039472	0.12446299	0.04371729	0.94701476
✓ Best GARCH model for Stock: GARCH(1,1) with BIC = 5.2256				

	mu	omega	alpha1	beta1
	0.009689856	0.006772600	0.023944047	0.968171056
✓ Best GARCH model for Gold: GARCH(1,1) with BIC = 2.6382				

Appendix B – ARMA Coefficients and BIC:

Best ARMA model for Bitcoin: ARMA(1,0) with BIC = 12507.54
Coefficients:

	ar1	intercept
	-0.0039	0.1830
s.e.	0.0217	0.0985

sigma^2 estimated as 20.79: log likelihood = -6242.27, aic = 12490.55

Best ARMA model for Stock: ARMA(1,0) with BIC = 11468.26
Coefficients:

	ar1	intercept
	-0.0103	0.1143
s.e.	0.0217	0.0767

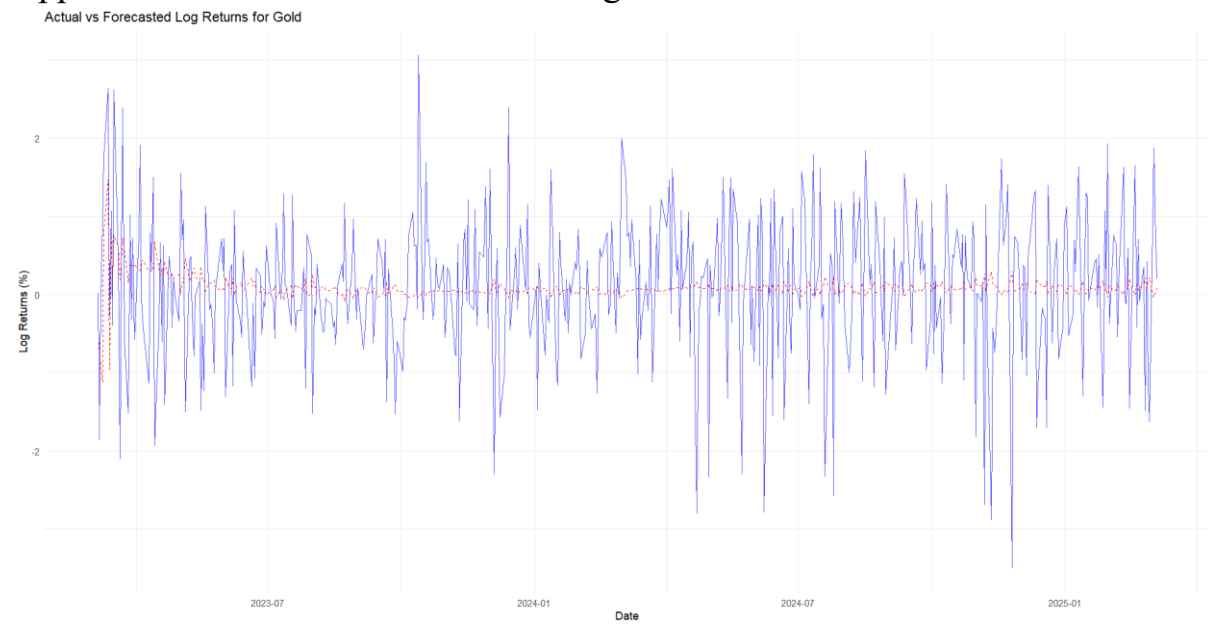
sigma^2 estimated as 12.75: log likelihood = -5722.64, aic = 11451.27

Best ARMA model for Gold: ARMA(0,1) with BIC = 5769.73
Coefficients:

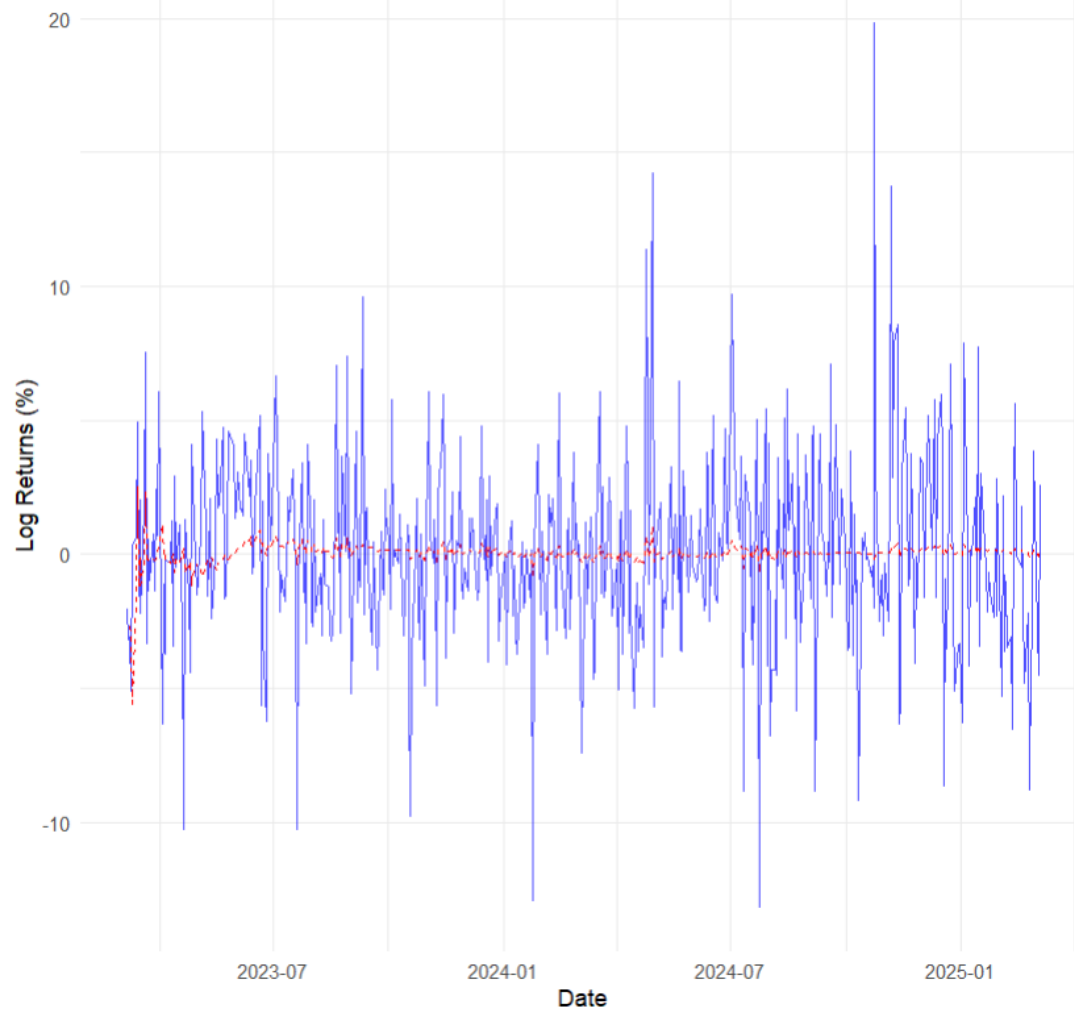
	ma1	intercept
	-0.0264	0.0190
s.e.	0.0217	0.0197

sigma^2 estimated as 0.8739: log likelihood = -2873.37, aic = 5752.75

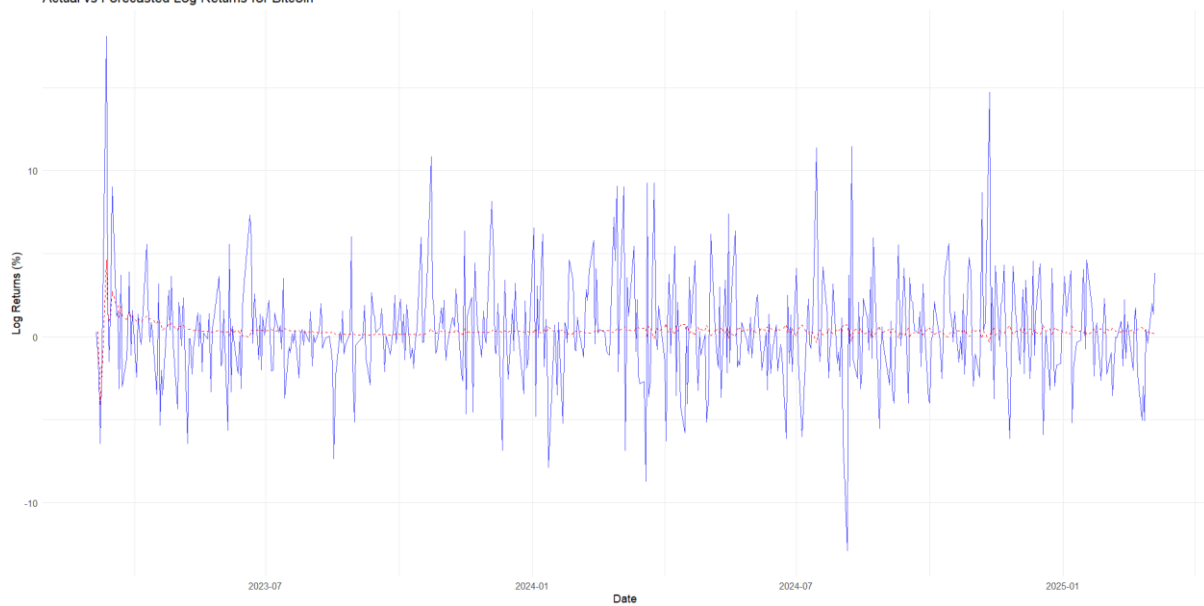
Appendix C – Actual vs. Forecasted Log Returns:



Actual vs Forecasted Log Returns for Stock



Actual vs Forecasted Log Returns for Bitcoin



Appendix D - Cumulative P&L for Gold and Tesla – 1-Day Straddle Strategy:

Cumulative P&L for Gold 1-Day Straddle Strategy



Cumulative P&L for TSLA 1-Day Straddle Strategy

